

DeGro: Target Determinacy Verification and Grounded Repair for Executable Mathematical Reasoning

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Abstract

Executable mathematical reasoning can succeed while encoding an underconstrained or mistranslated problem. Execution checks catch implementation failures but cannot establish whether the encoded constraints determine the requested quantity. We present *DeGro*, a neuro-symbolic framework for target determinacy diagnosis and grounded repair. DeGro asks whether two feasible assignments can disagree on the target; when they do, it accepts only source-supported constraints that pass provenance and formal checks, and otherwise abstains. We evaluate DeGro in controlled paired settings and in human-verified studies of naturally occurring formalization errors. Across the MIRA-Math and DRAW-Paired evaluations, DeGro generally improves repair-or-abstain accuracy over grounded self-review across multiple language models. On natural errors, it improves accuracy for supported MATH-500 cases, while ALG514 reveals answer-compatible formalization errors that ordinary answer matching can miss. Ambiguity detection remains substantially easier than repair, and explicit counterexample witnesses provide no measurable benefit beyond the determinacy verdict. Target determinacy is therefore an auditable guardrail for executable reasoning, not a guarantee of source-to-formal faithfulness.

CCS Concepts

• **Computing methodologies** → **Knowledge representation and reasoning**; **Intelligent agents**; *Natural language processing*.

Keywords

neuro-symbolic reasoning, mathematical agents, verification, counterexamples, abstention

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1 Introduction

Large language models increasingly delegate mathematical computation to programs and symbolic solvers. Program-aided reasoning reduces arithmetic errors, while systems such as SymCode generate self-contained symbolic programs and repair execution failures [5, 15]. Successful execution, however, does not establish that the encoded constraints determine the requested answer. A

program may run and return a value even when its formalization admits several values of the target.

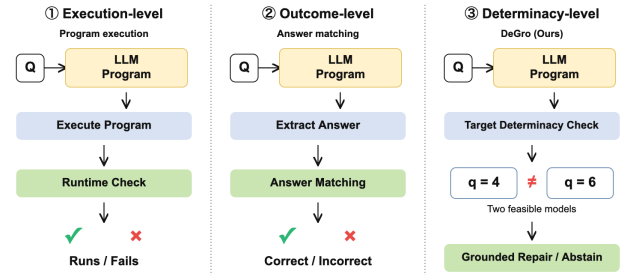


Figure 1: Three levels of executable mathematical reasoning evaluation. Execution checks whether a program runs; outcome evaluation checks its final answer. DeGro diagnoses target ambiguity through two feasible models with different target values (illustrated by $q = 4$ and $q = 6$), then attempts source-grounded repair with deterministic acceptance gates or abstains. Target determinacy is relative to the encoded specification and does not establish faithfulness to the source problem.

Consider: “Ten tasks are divided between A and B. A receives two more than B. How many does A receive?” A generated model containing only $a + b = 10$ is executable and satisfiable, but $(a, b) = (4, 6)$ and $(6, 4)$ imply different values of the target a . We call this condition *target ambiguity*: feasible assignments disagree on the requested quantity. It concerns the target rather than uniqueness of the entire assignment, since freedom in other variables need not affect the answer.

Detecting ambiguity does not establish how to resolve it. The same model faithfully represents the shorter source “Ten tasks are divided between A and B. How many does A receive?” For the complete source, adding $a = b + 2$ recovers an omitted relation; for the shorter source, the same addition fabricates information. The formalization and solver verdict are identical, but the correct action changes with source support. A repair policy must therefore distinguish evidence that the target is ambiguous from evidence that a particular constraint is justified.

We propose DeGro, a neuro-symbolic framework that separates these two decisions. It first constructs two renamed copies of a generated specification and asks whether both can satisfy the constraints while disagreeing on the target. A satisfying query provides auditable evidence of ambiguity. The agent then proposes a constraint with supporting source text, and acceptance checks require consistency, non-redundancy, and restored target determinacy. If



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no admissible repair is found, the agent abstains. The solver guarantee concerns the encoded specification: even a determinate target can be wrong when the target or constraints were mistranslated.

We evaluate this framework at two complementary levels. A controlled paired benchmark holds the formalization and diagnostic fixed while varying source support, testing whether the policy repairs a supported omission and abstains on its underspecified counterpart. Human-verified natural-error studies then examine representation coverage, ambiguity detection, and successful grounded repair separately. The controlled results show improved repair-or-abstain decisions, while the natural errors expose a substantial gap between detecting ambiguity and repairing its cause. We also test whether concrete witness assignments improve decisions beyond the solver verdict.

Our research questions are:

(RQ1) Decision quality and tradeoffs. How do target-determinacy diagnosis and source grounding affect repair-or-abstain accuracy and the tradeoff between supported repair and unsupported additions?

(RQ2) Paired source sensitivity. When the formalization and diagnostic are held fixed, can an agent both repair the source-supported omission and abstain on the underspecified member of the same pair?

(RQ3) Practical bottlenecks. How do representation coverage, ambiguity detection, and grounded repair limit effectiveness on natural errors, and do concrete witnesses improve controlled repair decisions beyond the verdict?

Takeaway

DeGro checks whether the encoded constraints determine the requested target, requires cited support and formal validation for repairs, and otherwise abstains.

Our main contributions are:

(1) A target-determinacy-guided repair-or-abstain framework. We introduce DeGro, which combines two-copy target-determinacy diagnosis with source-grounded repair, deterministic acceptance gates, and explicit abstention (RQ1).

(2) Controlled and natural-source paired benchmarks. We construct matched cases that hold the formalization fixed while varying whether the source supports the missing constraint. Alongside MIRA-Math, DRAW-Paired contributes 300 retained author-reviewed pairs derived from natural algebra word problems, isolating supported repair from unsupported completion (RQ2).

(3) A multi-stage empirical audit of executable reasoning. We combine controlled and human-verified natural-error evaluations to separate representation coverage, ambiguity detection, and grounded repair; we additionally test whether concrete witnesses improve decisions (RQ3).

The formal guarantee concerns F ; the empirical answers concern model and policy behavior.

2 Related Work

2.1 Executable Mathematical Reasoning

PAL and Program-of-Thoughts separate language-based reasoning from program execution to reduce computational errors [8, 15]. MathCoder and ToRA integrate code and tool calls into mathematical problem solving [18, 35], while faithful chain-of-thought connects generated rationales to executable symbolic procedures [25]. SymCode is particularly close to our setting: it represents mathematical reasoning as a SymPy program and uses execution feedback to repair programmatic failures [5]. These approaches motivate an additional check after successful execution. A generated artifact may return a value even though its constraints admit other values of the requested quantity. DeGro checks this property of the encoded feasible region and routes ambiguous specifications to grounded repair or abstention.

Outcome verification and process supervision assess candidate solutions or intermediate reasoning steps [10, 24, 33]. Self-Refine uses model-generated feedback, and CRITIC incorporates external tools into revision [17, 26]; evidence on intrinsic self-correction shows the limits of revision without reliable external feedback [20]. DeGro contributes a specific solver observation: two feasible assignments disagree on the target. This observation establishes ambiguity under the representation, independently of the predicted answer, but does not identify which source constraint should be added.

2.2 Symbolic Diagnosis and Relational Verification

Recent work addresses semantic failures that survive execution. ReLoop tests optimization formulations through solver-based parameter perturbations [22]. SymDiag translates reasoning steps into constraints and uses satisfiability and entailment checks to localize defects, with a Self-Auditor to distinguish translation errors from reasoning errors [11]. Minimal-core-guided repair instead returns an unsatisfiable subset of generated constraints as evidence for revising inconsistent translations [23, 30]. DeGro focuses on a complementary condition: the specification is satisfiable, but its feasible assignments disagree on the requested target. Its diagnosis concerns the specification as a whole rather than the validity of individual derivation steps, and its additive repair policy requires support from the original source.

The two-copy construction builds on self-composition and product programs, which reduce relational verification to queries over renamed copies [6, 7, 32]. DeGro instantiates this established technique as target agreement across feasible assignments. Agreement on the target is weaker than uniqueness of the full assignment: nuisance variables may remain free without making the answer ambiguous. Counterexample-guided refinement and synthesis likewise use solver evidence to guide changes [9, 31]. Here, however, a counterexample refutes target determinacy without supplying a semantically justified edit. Our contribution is the integration of this diagnosis with a source-sensitive repair policy and its evaluation, rather than a new relational verification primitive.

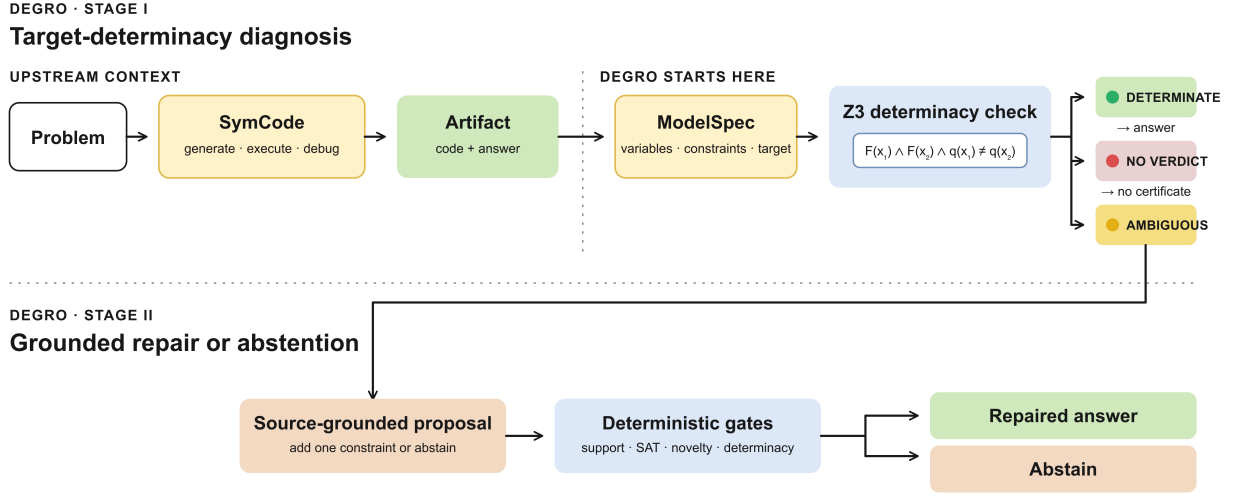


Figure 2: DeGro in the executable-reasoning pipeline. SymCode generates the executable artifact, while DeGro passes a structured MODELSPec to Z3-backed target-determinacy verification. A determinate target can be answered, an ambiguous target enters grounded repair and validation, and an inconclusive check returns no certificate.

2.3 Grounded Repair, Abstention, and Faithfulness

Classical diagnosis, abductive inference, and formal model repair provide foundations for identifying defects and modifying specifications to satisfy a desired property [3, 13, 29]. LLM-based repair extends this direction to declarative formal specifications [2]. Restoring determinacy alone is insufficient in our setting, since a guessed equality can force a unique target. DeGro therefore requests a source-supported constraint and checks consistency, non-redundancy, and restored determinacy before acceptance. These formal gates verify the effect of the edit; source support remains a separate semantic requirement.

Selective prediction studies when to withhold an answer under a risk-coverage tradeoff [14, 16], and semantic uncertainty estimates uncertainty in language-model outputs [21]. DeGro bases its repair-or-abstain decision on two distinct questions: whether the encoded target is determinate and whether the source supports a constraint that resolves ambiguity. MIRA-Math is directly relevant because it evaluates requesting and integrating necessary missing information [1]. We adapt its validated latent states into matched sources with identical formalizations but different support for repair, isolating the decision to recover an omitted relation or abstain from unsupported completion.

Autoformalization also separates formal validity from preservation of natural-language meaning [4, 36]. DeGro’s guarantee is similarly representation-relative: a determinate specification can still contain a wrong target or an unsupported constraint. Our paired and human-verified natural-error evaluations therefore distinguish determinacy diagnosis, grounded repair, and answer accuracy rather than treating a solver verdict as proof of source-to-formal faithfulness.

3 Methodology

DeGro checks whether a generated mathematical specification determines the requested target. Stage I diagnoses target ambiguity using a symbolic solver. Stage II attempts to resolve an ambiguous target by adding a constraint with source provenance, or abstains if no candidate is accepted. Figure 2 summarizes the pipeline.

3.1 Problem Setup and Target Determinacy

Given a natural-language problem P , the model produces a specification $F = (V, D, C, q)$, where V is the set of variables, D their domains, C the constraints, and $q(V)$ the requested expression. Its feasible assignments are

$$\text{Sol}(F) = \{m \mid m \models D \wedge C\}. \quad (1)$$

For a satisfiable F , the target is *determinate* when every feasible assignment gives the same target value:

$$\text{Det}(F, q) \iff \forall m_1, m_2 \in \text{Sol}(F), \quad q(m_1) = q(m_2). \quad (2)$$

Target determinacy does not require a unique assignment to all variables. For example, $x = 5, y \geq 0$ fixes the target $q = x$ even though y remains free. An inconsistent specification is reported separately and never receives a determinacy certificate.

The specification is stored as a typed MODELSPec: a structured record of variable sorts and bounds, source-linked constraints, and the target. Computational and verification backends consume this same representation, avoiding discrepancies between independently generated programs. The supported fragment includes integer, real, and Boolean variables, arithmetic equalities and inequalities, finite domains, and selected modular constraints.

The certificate concerns F , not whether F correctly represents P . Wrong or extra constraints can fix a target while mistranslating the source. DeGro therefore repairs only ambiguous specifications

by adding constraints; correcting a determinate but incorrect representation is outside its scope.

3.2 Stage I: Target-Determinacy Diagnosis

DeGro first checks whether $D \wedge C$ is satisfiable. An unsatisfiable result yields **INCONSISTENT**; a timeout or inconclusive solver result yields **UNKNOWN**. If the specification is satisfiable, DeGro constructs two copies with disjoint variable names and asks whether they can produce different target values:

$$\Phi(F, q) = F^{(1)} \wedge F^{(2)} \wedge q(V^{(1)}) \neq q(V^{(2)}). \quad (3)$$

If Φ is satisfiable, its projections form a *target-divergent witness pair*, and the status is **AMBIGUOUS**. If Φ is unsatisfiable, the status is **DETERMINATE**, since the initial check has already established that F has a solution. Unsupported representations yield **UNSUPPORTED** before solver verification. Only **AMBIGUOUS** enters Stage II.

Algorithm 1 Target-determinacy verification

Require: Typed specification $F = (V, D, C, q)$
Ensure: Status and optional witness pair
1: $r_0 \leftarrow \text{CHECKSAT}(D \wedge C)$
2: **if** $r_0 = \text{UNSAT}$ **return** **INCONSISTENT**
3: **if** $r_0 \neq \text{SAT}$ **return** **UNKNOWN**
4: $\Phi \leftarrow F^{(1)} \wedge F^{(2)} \wedge (q^{(1)} \neq q^{(2)})$ using disjoint copies
5: $(r_1, M) \leftarrow \text{CHECKSAT}(\Phi)$
6: **if** $r_1 = \text{UNSAT}$ **return** **DETERMINATE**
7: **if** $r_1 \neq \text{SAT}$ **return** **UNKNOWN**
8: **return** **AMBIGUOUS**, **PROJECTWITNESSES**(M)

Soundness. Assuming semantics-preserving compilation and a sound solver for the supported theory, $\text{SAT}(F)$ together with $\text{UNSAT}(\Phi)$ establishes target determinacy: any two feasible assignments with different targets would satisfy Eq. (3), contradicting unsatisfiability. Conversely, a satisfying assignment to Φ supplies two feasible witnesses with different targets. Timeout and **UNKNOWN** provide no certificate. The implementation uses a restricted expression compiler and separate variable maps for the two Z3 copies [12].

3.3 Stage II: Grounded Repair or Abstention

The verifier retains the target-divergent witness pair as a certificate, but the default repair prompt receives only the source problem, the encoded constraints and target, and the typed **AMBIGUOUS** verdict. It returns either **ADD-CONSTRAINT** with an exact supporting source span (or a permitted domain rule), or **ABSTAIN**. It receives neither the hidden gold target nor the missing gold constraint. An additional ablation exposes the concrete witnesses to test whether their values help beyond the verdict; they do not improve decisions on the present benchmark. In either variant, the model must recover a relation from the source rather than select a witness as correct.

Acceptance gate. For a proposed constraint c and a frozen set $\mathcal{R}$ of permitted domain-semantics rules, the deterministic gate accepts c only when

$$\text{Accept}(c) \iff \text{Compiles}(c) \wedge \text{Provenance}(c, P, \mathcal{R}) \wedge \text{SAT}(F \wedge c) \wedge \neg(F \models c) \wedge \text{Det}(F \wedge c, q). \quad (4)$$

Table 1: Controller semantics for DeGro integration.

Verifier state	Evidence	Controller action
Determinate	F sat, Φ unsat	answer with scope
Ambiguous	targets disagree	grounded repair
Inconsistent	F unsat	revise encoding
Unknown	solver inconclusive	fallback/escalate
Unsupported	outside fragment	alternate tool
No repair found	no accepted candidate	ask or abstain

These checks require that the candidate compile, cite a source span occurring in P or a rule in $\mathcal{R}$, preserve consistency, add a non-redundant restriction, and restore target determinacy. Compilation failures and inconclusive checks do not permit acceptance.

Provenance is an auditable check, not a proof that the constraint correctly expresses the cited text. Passing the gate therefore establishes valid provenance and determinacy under the repaired specification, rather than semantic correctness with respect to P . Semantic source support and functional correctness are evaluated separately for repair credit; an incorrect translation or unsupported target pin does not earn credit merely by passing the gate.

Bounded repair. DeGro allows at most two proposal attempts. The first accepted constraint is committed and terminates the loop. If the first proposal is rejected, the second attempt receives the rejection reason and operates on the unchanged F . Rejected constraints are not accumulated: each candidate must restore determinacy on its own, and a supported but insufficient addition is rejected. If the model abstains or neither candidate passes, the outcome is **NO-ADMISSIBLE-REPAIR-FOUND**, and F is preserved.

Failure to find an admissible repair does not prove that the source is underspecified: the model may have missed a relation already present in the source.

Controller policy. Table 1 maps verifier states to agent actions. A determinate result permits an answer scoped to the current specification, including an accepted repair when present. In interactive use, failure to find a repair can trigger clarification or abstention; the benchmark scores abstention because no clarification channel is available. Inconsistent, unknown, and unsupported states require distinct fallbacks rather than being treated as ambiguous inputs. The parser, compiler, solver, and gate form the trusted boundary; the language model only proposes specifications and repairs. Logs retain solver verdicts, witnesses, proposals, cited support, and rejection reasons for auditing.

4 Experiments and Results

4.1 Setup

4.1.1 Evaluation Scope. We evaluate repair policies along three complementary dimensions: (i) *diagnostic guidance*, whether a solver-produced target-ambiguity verdict improves revision; (ii) *source grounding*, whether exact source evidence prevents unsupported additions; and (iii) *repair-or-abstain behavior*, whether a policy repairs a supported omission while withholding an edit when the source is genuinely underspecified. The first two

dimensions address RQ1, whereas repair-or-abstain behavior on matched sources directly addresses RQ2.

4.1.2 Baselines. To evaluate the effectiveness of DeGro, we compare it against representative repair policies that use weaker diagnostic or grounding signals. All methods receive the same source, current constraints, and target; begin from the same initial formalization; and use a two-attempt repair budget. Differences therefore reflect the feedback and repair policy rather than initial-generation quality. We consider the following baselines:

- **Self-review.** The model revises the formalization using the source and current artifact, but receives no explicit solver diagnosis and is not required to ground an added constraint in the source.
- **Grounded self-review.** The model follows the same exact-span grounding rule used by DeGro, so every introduced constraint must cite supporting source text. It receives no solver feedback, isolating grounding without determinacy guidance.
- **Determinacy-guided repair.** The model receives the typed `AMBIGUOUS` verdict but no exact-span grounding requirement. This baseline isolates the effect of determinacy feedback without source-grounded control.

4.1.3 Models. We evaluate all four policies with GPT-OSS 20B and GPT-OSS 120B [28], and Nemotron 3 Nano 30B [27]. All models use the same cases, frozen prompt, temperature 1.0, and repair budget.

4.1.4 Metrics. Accuracy. Our primary metric counts a decision as correct when the method produces a grounded, functionally correct repair for an omission case or abstains on an underspecified case. Because each controlled pair contains one case of each type, accuracy is equivalently computed as

$$\text{Accuracy} = \frac{\text{Recall} + \text{Specificity}}{2}.$$

Recall and specificity. Recall measures the proportion of omission cases repaired correctly, whereas specificity measures the proportion of underspecified cases on which the method correctly abstains. We omit the false-positive rate because it is exactly $1 - \text{Specificity}$. **Unsupported repair rate.** To assess grounding, we report URR as the proportion of proposed additions that lack source support. URR is interpreted alongside recall, since a policy that always abstains would produce no unsupported additions but provide no repair utility. Pair-cluster confidence intervals preserve within-pair dependence, and semantic, functional, and grounded outcomes are evaluated separately.

4.2 Datasets

We evaluate on two matched-pair benchmarks. MIRA-Math provides controlled source variations across six mathematical families, while DRAW-Paired transfers the same repair-or-abstain design to natural algebra word problems. Figure 3 summarizes their composition.

4.2.1 MIRA-Math: Controlled Sources. Construction. For each gold specification $F^* = \{c_1, \dots, c_k, c^*\}$, we remove one target-critical constraint to obtain $F^- = F^* \setminus \{c^*\}$. We retain the item only when F^* determines the target and F^- remains satisfiable but

Paired benchmark composition

660 pairs across two benchmarks

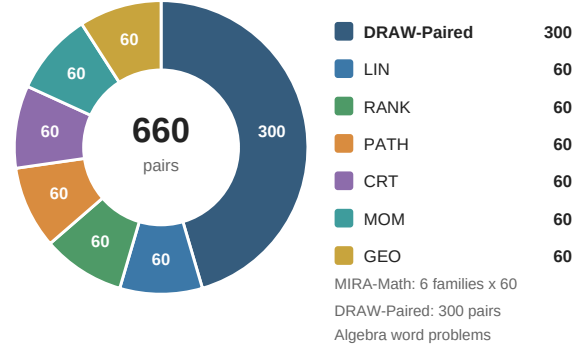


Figure 3: Paired benchmark composition. DRAW-Paired contributes 300 pairs constructed from natural algebra word-problem sources; MIRA-Math contributes six 60-pair families: linear separation (LIN), rank-deficient systems (RANK), path sums (PATH), Chinese-remainder reconstruction (CRT), moment constraints (MOM), and geometry (GEO). Every pair yields one `OMISSION` and one `UNDERSPECIFIED` case.

target-ambiguous. Both members of a pair share F^- : the `OMISSION` source retains the fact represented by c^* and should be repaired, whereas the `UNDERSPECIFIED` source removes that fact and should trigger abstention. Thus the formalization and solver verdict are fixed, and only source support reveals the correct action.

Composition and validation. MIRA-Math contributes 360 pairs (720 cases) across six families: linear separation, rank-deficient systems, path sums, Chinese-remainder reconstruction, moment constraints, and coordinate geometry [1]. Each family contains 60 pairs. The builder imports the typed missing fact supplied by the dataset, rather than asking an LLM to corrupt the problem, and every retained deletion passes satisfiability, ambiguity, and restored-determinacy checks. The manifest records source revision 72ff918, license, selection seed, family counts, and prompt/output hashes. The supplement reports the frozen 120-pair held-out split and Geometry60 separately.

4.2.2 DRAW-Paired: Natural Sources. Construction. DRAW-Paired is derived from DRAW-1K’s 600 training and 200 development problems [34]; the 200 test problems remain untouched. After removing one duplicate identifier and 200 single-equation problems, 599 systems are solver-eligible. For each retained source, deleting one gold equation preserves satisfiability but makes the real-valued target ambiguous, while the full system recovers a published gold solution. Pair members share the retained equations, target, and domains; only the supporting source span is removed from the underspecified member. These are controlled omissions applied to natural sources, not naturally occurring model errors.

Selection and review. Gold coefficient alignments identify candidate supporting sentences. The final set contains 300 pairs (600 cases). All retained spans and deletions were author-reviewed after inference; a subsequent re-review removed 50 candidate pairs, and

the released exclusion list records every removed identifier. Two title-abbreviation errors were corrected and the affected pairs rerun. The supplement provides the full selection and review provenance.

4.3 Main Paired Experiment on MIRA-Math

Primary comparison. We compare DeGro with grounded self-review on accuracy using exact paired McNemar tests and Holm correction across three models. Percentage-point differences use 10,000 pair-cluster bootstrap resamples. The 360-pair comparison includes method-development observations and the geometry extension added after inspection of the original results; the supplement retains the 120-pair held-out results and original analysis protocol. GPT-OSS supports configurable reasoning effort and structured outputs [28].

4.4 Main Results

RQ1: repair-or-abstain decisions. Table 2 reports the 360-pair MIRA-Math aggregate. DeGro raises accuracy from 91.7% to 96.3% for GPT-OSS 20B (+4.58 points, 95% CI [2.08, 7.08]; 56 versus 23 discordances; exact $p = .000264$) and from 89.6% to 96.1% for 120B (+6.53 points, CI [4.31, 8.75]; 61 versus 14; $p = 3.81 \times 10^{-8}$). Nano changes from 86.7% to 89.6% (+2.92 points, CI [0.42, 5.42]; 54 versus 33; $p = .0314$). Holm-adjusted p -values are .000527, 1.14×10^{-7} , and .0314. DeGro has the highest accuracy among the four policies for each model; Geometry60 was added after inspection of the original cohort, so this combined result requires confirmation on an independently frozen evaluation.

RQ1: source-grounded acceptance. On MIRA-360, 20B DeGro reduces URR from 11.2% to 2.9% and raises specificity from 88.6% to 99.2%, while recall falls from 94.7% to 93.3%. For 120B, URR falls from 15.5% to 6.4%, specificity rises from 82.5% to 94.4%, and recall from 96.7% to 97.8%. Nano reaches 99.7% specificity and 1.0% URR, versus 98.3% and 4.6%, while recall rises from 75.0% to 79.4%. These metrics expose model-specific repair–abstention tradeoffs. Figure 5 illustrates rejection of a superficially grounded but non-resolving edit.

4.5 Main Natural-Source Experiment: DRAW-Paired

Table 2’s right-hand block reports the retained DRAW-Paired results after the author’s post-inference exclusion of 50 pairs. Relative to grounded self-review, DeGro improves accuracy by 5.8 points for 20B, 8.8 for 120B, and 10.2 for Nano. All three improve abstention and supported repair; 20B recall rises from 62.3% to 66.7%. DeGro and verdict-only repair tie at 77.8% accuracy for 120B: verdict-only repair has higher recall, while DeGro has higher specificity. Exact-span provenance does not establish semantic entailment, and high URR requires reporting repair utility alongside abstention. Because exclusions followed access to outcomes, these estimates describe the retained subset and do not establish an independent replication.

GPT-OSS 20B improves by +5.83 accuracy points (pair-cluster 95% CI [2.5, 9.0]; exact McNemar $p = .00049$; Holm-adjusted $p = .00049$), with 66 versus 31 discordant cases favoring DeGro versus grounded self-review; GPT-OSS 120B improves by +8.83 accuracy points (pair-cluster 95% CI [5.7, 12.2]; exact McNemar $p = 6.07 \times$

10^{-8} ; Holm-adjusted $p = 1.82 \times 10^{-7}$), with 75 versus 22 discordant cases favoring DeGro versus grounded self-review; Nano improves by +10.17 accuracy points (pair-cluster 95% CI [6.0, 14.5]; exact McNemar $p = 1.72 \times 10^{-6}$; Holm-adjusted $p = 3.44 \times 10^{-6}$), with 111 versus 50 discordant cases favoring DeGro versus grounded self-review. These tests use 10,000 pair-cluster bootstrap resamples and Holm adjustment across three model-specific comparisons. All three retained-subset comparisons are significant after adjustment, but exclusions occurred after inference and access to outcomes. These results do not establish a preregistered replication or a model-universal benefit.

5 Analysis

5.1 Additional and Natural-Error Analysis

Held-out comparison. The supplement retains the 120-pair results separately: DeGro minus grounded self-review is +5.0 accuracy points for 20B (CI [0.4, 9.6]), +6.25 for 120B (CI [1.7, 10.8]), and +1.25 for Nano (CI [−2.5, 5.0]). The direction agrees with the 300-pair results, but neither Nano estimate establishes an incremental accuracy benefit. The main comparison does not replace an independent held-out evaluation.

Source-coverage prompt. Error analysis on the 80-pair development set motivated one frozen prompt refinement: explicitly compare every source condition with F , distinguish ambiguity of F from missing information in the source, and expose already enforced variable domains. The 300 solver-compatible MIRA pairs were already exhausted, so we tested this refinement on a new 100-pair synthetic stress set spanning four algebraic families; this is neither an external benchmark nor an additional MIRA test set. We concurrently reran the original DeGro prompt. For Nano 30B, accuracy rises from 75.5% to 95.0% (+19.5 points, pair-cluster 95% CI [13.5, 25.5]; exact $p < .001$), with recall rising from 52.0% to 90.0% and specificity from 99.0% to 100.0%. This increased repair activity also raises URR from 1/53 (1.9%) to 8/98 (8.2%). For GPT-OSS 120B, accuracy changes from 98.5% to 99.5% (+1.0 point, CI [−1.0, 3.0]; $p = .625$), while URR falls from 3/102 (2.9%) to 1/101 (1.0%). Holm adjustment over these two tests gives $p < .001$ and $p = .625$. The supplement reports the full protocol and table; these results motivate a future frozen external evaluation and do not revise the main comparisons above.

5.1.1 Natural-Error Evaluation Protocol. Cohort construction.

To evaluate naturally occurring formalization errors, GPT-OSS 20B generates one Structured SymCode specification for each MATH-500 problem without injected corruption. We freeze all parsed, solver-supported, conclusive cases before annotation or outcome inspection and reuse each initial specification across methods. MATH-500 contributes 292 of 500 conclusive formalizations, while ALG514 contributes 503 of 514; exclusions are reported as coverage outcomes.

Human verification. Two reviewers inspect the final labels for all 292 MATH-500 and 503 ALG514 cases. Labels distinguish correct, missing/wrong/extra constraint, wrong domain, wrong target, computational error, and other; only source-supported, target-critical omissions enter the primary subset. For MATH-500, GPT-OSS 120B subsequently produces a fresh blinded proposal for every case and

Table 2: Main paired experiments on MIRA-360 (360 pairs, 720 cases across six families) and DRAW-Paired (300 retained author-reviewed pairs, 600 cases). Values are percentages. Accuracy is primary; recall and specificity decompose repair and abstention, and URR measures unsupported additions among proposed additions. Blue rows denote DeGro; bold accuracy marks the best policy within each model and dataset. DRAW-Paired review followed inference, and its extension was selected after initial outcomes.

Model	Method	MIRA-Math				DRAW-Paired			
		Acc.↑	Recall↑	Spec.↑	URR↓	Acc.↑	Recall↑	Spec.↑	URR↓
GPT-OSS 20B	Self-review	90.6	93.1	88.1	14.5	73.8	66.7	81.0	43.3
	Grounded self-review	91.7	94.7	88.6	11.2	73.3	62.3	84.3	43.8
	Determinacy-guided	92.6	88.1	97.2	7.9	73.7	57.0	90.3	46.1
	DeGro (ours)	96.3	93.3	99.2	2.9	79.2	66.7	91.7	36.3
GPT-OSS 120B	Self-review	78.6	99.4	57.8	29.9	72.0	68.3	75.7	44.1
	Grounded self-review	89.6	96.7	82.5	15.5	69.0	54.0	84.0	51.1
	Determinacy-guided	92.4	98.6	86.1	13.4	77.8	69.3	86.3	37.5
	DeGro (ours)	96.1	97.8	94.4	6.4	77.8	63.7	92.0	38.8
Nemotron	Self-review	83.9	76.1	91.7	25.5	63.5	44.7	82.3	59.0
Nano 30B	Grounded self-review	86.7	75.0	98.3	4.6	60.7	35.7	85.7	61.9
	Determinacy-guided	72.9	53.9	91.9	27.3	63.3	34.7	92.0	55.4
	DeGro (ours)	89.6	79.4	99.7	1.0	70.8	48.3	93.3	44.4

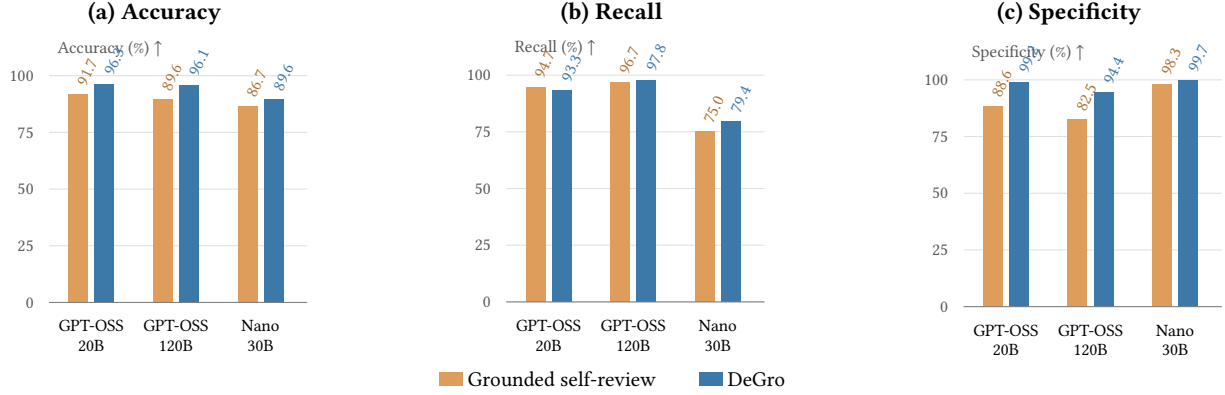


Figure 4: MIRA-360 results. DeGro has higher accuracy than grounded self-review for all three models after Holm correction. Recall and specificity expose model-specific tradeoffs. All axes start at zero; the aggregate includes pilot and development observations plus Geometry60, which was added after the original outcomes were inspected.

agrees with the human-verified primary label on 207 of 292 cases (70.9%), illustrating why model proposals are not treated as ground truth. For ALG514, reviewers inspect all final labels after GPT-OSS 120B proposes 503 labels and re-adjudicates the 22 cases in which semantic status disagrees with verifier-compatible numerical status. Because separate pre-discussion labels were not retained, we claim human verification rather than independent annotation or inter-annotator agreement.

Evaluation scope. The original problem is the unit of analysis, and each method reuses the same initial formalization. The primary measures are conclusive solver coverage, ambiguity recall, and end-to-end grounded repair; secondary measures include precision, false ambiguity or repair, and preservation of correct controls. Conditional repair counts accepted grounded fixes among detected

omissions, whereas end-to-end repair retains every verified omission in the denominator. Proportions use Wilson 95% intervals, and paired answer accuracy uses exact McNemar tests. Numeric agreement alone does not count as repair success. The original DRAW-1K natural-error audit remains omitted because corrected-verifier outcomes and final human labels were incomplete; it is distinct from the gold-equation-derived DRAW-Paired experiment.

RQ3: human-verified natural errors. On the supported MATH-500 cohort, 158 of 292 formalizations are correct; the remaining labels are 69 wrong-target, 32 wrong-constraint, 24 target-critical missing-constraint, eight extra-constraint, and one wrong-domain. The detector identifies 21 of 24 target-critical omissions (87.5% recall, 95% Wilson CI [69.0, 95.7]) but precision is 17.4% (21/121), and only one omission is successfully repaired end to end (1/24 =

Query and current formalization			
Source: $-2a + 3b - 4c = 15$, $-a + 2b + 4c = -8$. Find the integer value of a . Encoded constraints: the two displayed equations; all variables are already integers.			Solver verdict: AMBIGUOUS.
Self-Review	Grounded Self-Review	Determinacy-Guided	DeGro (ours)
Model output ADD_CONSTRAINT Cites: "Find the integer value of a ." $a = \text{int}(a)$ Redundant with the declared domain; the target remains ambiguous. ✗	Model output ADD_CONSTRAINT Cites: "integer value of a " $a = \text{int}(a)$ Lexically grounded, but non-resolving and already enforced. ✗	Model output ADD_CONSTRAINT Cites: "Find the integer value of a ." $a \bmod 1 = 0$ Reacts to ambiguity, but invents no missing source relation. ✗	Verified decision ABSTAIN Ambiguous target; no unused source relation supports a repair. No constraint added Correctly withholds an unsupported completion. ✓

Figure 5: Illustrative GPT-OSS 120B output on MIRA-Math case 1in-000271. The source supplies two equations for three integer variables, so the target is not uniquely determined. All three baselines add an integrality condition that is already enforced and does not resolve ambiguity; DeGro is the only method to abstain. This selected example explains the mechanism and is not a substitute for the aggregate comparison in Table 2.

4.2%). DeGro nevertheless raises cohort answer accuracy from 49.3% (144/292) to 53.8% (157/292); 16 discordant cases favor DeGro and three favor the structured baseline (exact McNemar $p = .0044$). This prevalence is conditional on 58.4% solver-fragment coverage.

ALG514 [34] provides a higher-coverage contrast: 503 of 514 cases enter the verified cohort, of which 477 are correct, 11 wrong-target, eight missing-constraint, and seven wrong-constraint. The semantic-error rate is 5.2% (95% Wilson CI [3.6, 7.5]). Structured Solver and DeGro score 94.2% (484/514) and 94.4% (485/514), respectively, using order-invariant subset matching with 10^{-3} tolerance for ALG514’s all-unknown, rounded solution vectors. Nine numerically compatible answers retain a human-verified semantic error, directly showing that answer-only evaluation can conceal representation failures. The one-case DeGro gain on ALG514 is descriptive and requires independent confirmation.

RQ3: detection–repair decomposition. The RQ3 results separate three stages that an aggregate answer score conflates. First, representation coverage asks whether the structured language and solver can express and decide the generated model. Second, ambiguity detection asks whether target-divergent assignments exist within that model. Third, grounded repair asks whether the original source supports a new constraint that removes the disagreement. MATH-500 performs moderately on the first stage (292 of 500), strongly on recall at the second (21 of 24 omissions), and poorly on the third (one of 24). Thus RQ3 identifies conversion of a solver counterexample into a semantically correct, cited edit as the main bottleneck on this cohort.

Precision provides the complementary view. Only 21 of 121 ambiguous MATH-500 specifications are verified target-critical omissions. The remaining ambiguous cases include wrong targets, wrong constraints, and representations whose feasible set is broad for reasons other than a missing source fact. These are legitimate

warnings about F but not opportunities for additive repair. A high-recall detector can consequently be useful for routing while still requiring a strict gate: automatically treating every ambiguity as an omission would reproduce the over-repair behavior observed in generic self-review.

Answer accuracy alone understates the semantic distinction. On MATH-500, DeGro changes the shared structured baseline from 49.3% to 53.8%, with 16 discordances favoring DeGro and three favoring the baseline. Yet only one verified omission is repaired end to end, so most accuracy changes do not constitute evidence that the missing-relation mechanism was solved. On ALG514, nine numerically compatible outputs still carry a verified semantic defect. We therefore report numerical, semantic, functional, grounding, and determinacy outcomes separately rather than reinterpret every answer improvement as representation repair.

Decision-theoretic reading. Accuracy assigns equal mass to repair on complete sources and abstention on underspecified sources because every controlled item contributes one of each. Real applications may penalize fabrication more heavily than missed repair. The case-level release permits alternative utilities, but the qualitative tradeoff is already visible: generic grounded review attains high recall while over-repairing, whereas a bare determinacy verdict favors abstention. DeGro attempts to recover useful repairs without surrendering the conservative action boundary. Whether that balance is preferable depends on the downstream cost of an unsupported assertion, which should be declared rather than hidden inside a single accuracy metric.

5.2 Witness Ablation and Family Effects

RQ3: witness ablation. On the 100-pair development set, DeGro scores 98.5% accuracy versus 97.5% with witnesses (five versus three discordances favoring DeGro; Holm-adjusted $p = .727$). A minimal witness raises recall by six points over a bare verdict, also

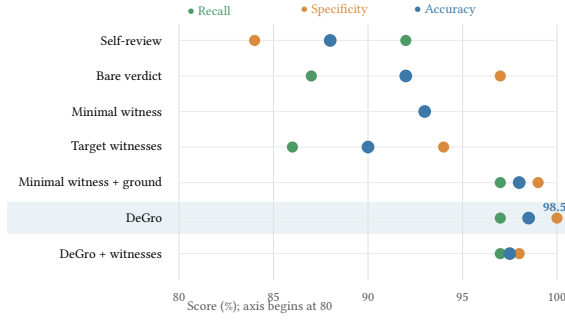


Figure 6: GPT-OSS 20B ablation on 100 development pairs. Concrete witnesses do not improve accuracy over DeGro without witnesses. The false-positive rate is omitted because it is exactly 100 – Specificity.

non-significantly ($p = .714$). Concrete assignments therefore add no measurable benefit here: the typed verdict can be exposed by default and witnesses retained for audit or harder cases.

This part of RQ3 is answered only on the 100-pair controlled development set. Every instance contains one target-critical omission, so non-uniqueness already localizes the relevant defect; witness assignments add tokens but little new information. Specifications with several missing facts or many nuisance dimensions may differ, but these data support determinacy-guided grounding, not a universal benefit from exposing concrete assignments.

RQ1: unsupported additions and repair tradeoffs. For 20B, a bare verdict raises specificity to 98.0% while lowering recall to 87.7%; DeGro restores repairs, reaches 99.0% specificity, and reduces unsupported additions from 12.5% to 3.4% relative to grounded self-review. For GPT-OSS 120B, DeGro raises specificity from 79.7% to 93.3% and recall from 97.0% to 98.0%, while reducing unsupported additions from 17.3% to 7.0%. Family effects remain concentrated rather than uniform.

Figure 7 explains the RQ1 tradeoff behind the aggregate scores. Generic review often inserts a plausible but unstated relation; DeGro blocks many such additions. The 20B rank-deficient family remains an adverse regime, whereas 120B improves in all five families. Lower URR therefore cannot by itself establish better grounded decision-making; recall, specificity, and family-specific losses must be read with it.

5.3 Coverage Analysis

RQ3: broad-benchmark coverage audit. To measure the representation bottleneck preceding verification, GPT-OSS 20B generated one shared Structured SymCode specification for each MATH-500 problem [19]. Of 500 specifications, 292 (58.4%) parsed, reached the solver, and produced a conclusive verdict; Figure 8 accounts for every exclusion.

Coverage is fixed solely from parsing and solver state, not observed answers. Solver-level unsupported cases dominate and require a broader theory or compiler; model-level failures instead require better structured generation. Conditional accuracy concerns the 292 supported problems, whereas end-to-end accuracy assigns

failures to the other 208. All Structured Solver records completed; DeGro completed 285, with six call errors and one unsupported record. We retain the frozen denominator and count all seven as incorrect.

This separation prevents selection of a favorable coverage slice after outcomes are known. Conditional accuracy asks how well the agent performs once a specification reaches a conclusive solver state; end-to-end accuracy additionally charges every representation and execution failure. Reporting only the former would make a narrow verifier appear competitive with unrestricted program generation while hiding 41.6% of the benchmark. The exclusion categories also route improvements to the right component: unsupported solver constructs call for broader compilation, while failures before a valid MODELSPec call for better structured generation.

5.4 Discussion and Limitations

Interpretation and validity. The paired design holds latent values, target, domains, retained constraints, and solver verdict fixed; only the source support for c^* changes. The repair policy never receives c^* , the gold target, or the gold action. The 120 held-out pairs were frozen before inspection. The original 300-pair evaluation includes pilot and development observations used during method development, and the 60-pair geometry extension was added after inspecting those outcomes; consequently, MIRA-360 is not an independent held-out test. The witness ablation uses development data, and the source-coverage refinement was designed from development errors and tested only on synthetic instances from the same task templates; its large Nano gain requires confirmation on an independently sourced frozen benchmark. Accuracy equally weights repair and correct abstention; URR must be interpreted with recall and specificity because always abstaining produces zero unsupported additions but no repair utility.

A conventional mixture of complete and incomplete questions admits shortcuts: families may differ in difficulty, wording may reveal that a fact is missing, and a policy may simply learn to repair every ambiguous solver state. Our paired construction rules out these shortcuts because both members share the same formalization and diagnostic. Correct behavior changes only with the natural-language support for the withheld constraint. Exact McNemar tests then depend only on pairs for which the two policies disagree, while pair-cluster bootstrap intervals preserve the dependence between the two case types.

The primary contrast also controls the language-model interface. Both methods receive the same source, specification, output schema, grounding instruction, model, and token budget; the added determinacy signal is the intended treatment. Gold latent values are used only after generation for scoring. A direct target pin is considered unsupported unless a source span licenses it and all semantic, functional, and determinacy gates accept it. Retaining raw and gated outputs reveals when apparent accuracy is purchased by fabrication.

Scope of the claim. The evidence supports target determinacy as a guard against plausible but unstated repairs, including a five-point GPT-OSS gain. It does not establish semantic equivalence between P and F , a universal grounding benefit, an incremental witness benefit, or superiority over unrestricted code generation. Exact

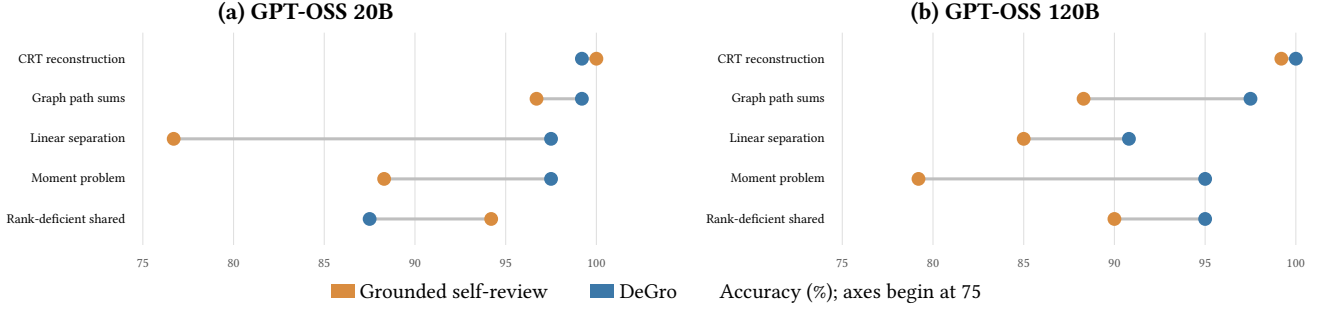


Figure 7: Accuracy by family on the original five-family MIRA-300 cohort; each family contains 60 pairs. DeGro improves four of five families for GPT-OSS 20B and all five for GPT-OSS 120B. Geometry60 is reported separately in the supplement.

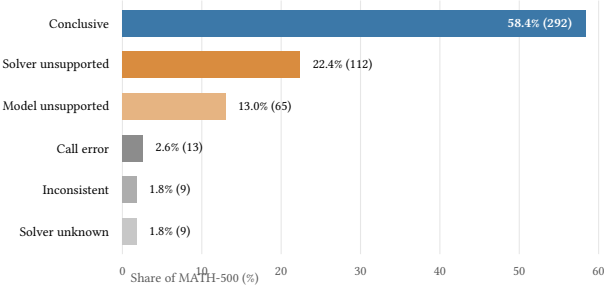


Figure 8: Frozen MATH-500 cohort accounting for all 500 shared initial structured specifications. Solver-level unsupported cases are the largest exclusion category.

spans are auditable provenance rather than semantic entailment. Confidence and determinacy are complementary: low confidence may justify fallback for a determinate target, while confidence cannot authorize an ambiguous one.

The distinction is important for agent design. An execution monitor answers whether an artifact runs; an outcome verifier may check a final answer; DeGro instead characterizes the feasible set before the agent commits to a value. This intermediate certificate supports selective action: answer when the target is determined, repair only when a cited source fact restores determinacy, and otherwise ask, abstain, or escalate. It is not a proof that the formalization matches the source. A correctly solved but incorrectly translated model may remain determinate.

Grounding itself is also only a proxy. A model can cite the correct sentence yet encode the wrong equation, while an implicit mathematical convention may justify an expression without a contiguous span. We therefore keep semantic validity, functional validity, provenance, and solver determinacy as separate fields. This decomposition makes disagreement auditable and prevents numerical answer correctness from silently absorbing translation errors.

External validity. MIRA cases are controlled but templated; Geometry60 uses an adapter-specific squared-distance target and was added after the original results were inspected. MATH-500 is natural but only 58.4% lies inside the present solver fragment, whereas ALG514 reaches 97.9% coverage in a narrower domain. Labels were human-verified after model proposals, not independently annotated,

and dataset prevalence is therefore reported separately. Natural errors contain only 24 MATH-500 and eight ALG514 target-critical omissions: detection is high but repair remains limited. The two GPT-OSS sizes also differ materially in baseline abstention behavior, bounding any size-general claim.

The controlled benchmark deliberately removes one known fact and therefore estimates a clean repair-versus-abstain decision under idealized representation control. It underrepresents paraphrase, irrelevant prose, diagrams, implicit conventions, wrong constraints, and interacting translation errors. The natural cohorts contain these phenomena but trade away causal control. Their purposes are complementary: the paired benchmark identifies a policy effect, while the natural study estimates what the current pipeline actually encounters. We do not merge them into one leaderboard number.

The annotation protocol also avoids defining positives with DeGro’s own verdict. Detector outputs, repairs, method names, and downstream answer outcomes are hidden from the initial packet. A target-critical omission must instead be established from the source and generated specification; correct, wrong-constraint, wrong-target, and wrong-domain cases remain in the denominator. Because model proposals seeded review and no separate blinded pre-discussion labels were preserved, we claim human verification rather than independent annotation or inter-annotator reliability.

Reproducibility and risk. The artifact includes the compiler, two-copy query, gates, tests, builders, frozen manifests, prompts, raw records, and analysis, including both additional Nemotron runs. Each MIRA-360 analysis consumes 2,880 records per model; the supplement retains the 120-pair outputs and their complete hashes. Determinacy remains relative to F : wrong or extra constraints may force a unique wrong target, non-target omissions are invisible, and the current SMT fragment excludes much nonlinear, transcendental, and diagrammatic reasoning. Interfaces must expose this scope rather than imply truth. The data contain no personal information; AI assistance requires venue-compliant disclosure and does not reduce author responsibility.

The manifests record dataset revision and license, selection seed, family counts, exact model strings, decoding settings, prompt hashes, timestamps, and full output hashes. Analysis scripts reconstruct every table from JSONL, reject incomplete expected pair counts, and preserve whether a row came from pilot, development, held-out, or geometry-extension data. Natural-error packets and

verified labels are separately blinded and content-addressed. These records reproduce the MIRA-360 main table, its original 300-pair component, and the separate 120-pair held-out comparison.

Several failure modes remain outside the certificate. A wrong or extra constraint may force a unique but incorrect target; an omitted fact that does not affect q is invisible; several interacting omissions may exceed the bounded repair budget; and witnesses may differ only on irrelevant variables. The supported SMT fragment excludes much nonlinear, transcendental, and diagrammatic mathematics. These are reasons to surface the encoded constraints, cited repair, and solver status in the interface, not to display an unqualified “verified” badge.

Finally, the tested models exhibit different instruction-following and abstention behavior, and hosted checkpoints may change. Nano’s original 300-pair comparison was inconclusive; its significant MIRA-360 difference appears only after the geometry extension was added and therefore requires independent replication. DeGro’s advantage should be treated as a property of a specified model–prompt–gate configuration. The GPT-OSS 120B baseline over-repairs many underspecified cases despite its larger size. Deterministic gates remain necessary, and deployment thresholds should reflect the relative cost of fabricated facts and missed repairs.

Deployment implications. DeGro is best used as a routing signal rather than a universal correctness badge. A determinate, supported specification can proceed to answer generation; an ambiguous specification with a grounded candidate can enter the bounded repair loop; and an ambiguous specification without support should trigger clarification or abstention. Inconsistent, unknown, and unsupported cases require different fallbacks, as Table 1 makes explicit. Keeping these states typed prevents an agent from treating lack of a solver certificate as permission to guess.

The results also suggest a conservative interface. The system should expose the target expression, the relevant encoded constraints, the determinacy status, and any source span used for repair. When a witness exists, it can be retained in the audit record even if the default language-model prompt receives only the verdict. This design follows the ablation result: witnesses are valuable mechanical evidence, but displaying them to the repair model did not improve decisions on the present benchmark. Users can then distinguish “the current model fixes this quantity” from the stronger and unsupported claim that the entire translation is correct.

Deployment comparisons should report representation coverage, decision quality conditional on a conclusive verifier state, and end-to-end utility with unsupported, unknown, and call-error cases retained as failures. Otherwise, a narrow symbolic verifier can appear reliable by declining difficult inputs before scoring.

6 Conclusion

DRAW-Paired extends the design to 300 retained author-reviewed natural algebra sources. DeGro improves accuracy over grounded self-review across the three models, and ties verdict-only repair for GPT-OSS 120B. Both cohorts were reviewed after inference, the extension was selected after initial outcomes, and 50 pairs were subsequently excluded after re-review; these results complement the controlled MIRA-Math evaluation but require confirmation on

data frozen before inference. On MIRA-360, DeGro raises accuracy over grounded self-review by 4.58 points for GPT-OSS 20B, 6.53 for 120B, and 2.92 for Nano, with all three comparisons surviving Holm correction. The combined set includes pilot and development observations plus a 60-pair geometry extension added after the original outcomes were inspected; Nano’s original 300-pair difference was inconclusive, so the model-general pattern requires confirmation on an independently frozen benchmark. Separate 120-pair held-out results remain in the supplement. For RQ2, decisions depend on source support even when formalization and determinacy signal are identical. For RQ3, coverage and detection substantially exceed end-to-end grounded repair on natural errors, while concrete witnesses add no measurable benefit on the single-omission development set. Target determinacy is a representation-relative routing condition, with grounding and coverage reported alongside decision accuracy.

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